

# EXACT ORNSTEIN-UHLENBECK NUMERICAL UPDATE RULE

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ABSTRACT. In this supplement, we derive the exact numerical update rule used in simulating the Ornstein-Uhlenbeck magnetic field model, justifying its usage as opposed to the Euler-Maruyama scheme.

Let  $X_t$  be a random variable satisfying the Ornstein-Uhlenbeck SDE

$$dX_t = -\lambda X_t dt + \sigma dW_t, \quad \lambda = \frac{1}{\tau_c} > 0.$$

We wish to derive an exact discrete-time update for  $X_{n+1} = X(t_n + \Delta t)$  given  $X_n = X(t_n)$ .

## 1. EXISTENCE AND FORM OF THE SOLUTION

The SDE is linear with globally Lipschitz drift  $b(x) = -\lambda x$  and constant diffusion coefficient  $\sigma(x) = \sigma$ . Therefore, by Itô's existence and uniqueness theorem, there exists a unique strong solution to the SDE.

**Theorem 1.** (*Itô's product rule*) *If  $X_t$  and  $Y_t$  are semimartingales, then*

$$d(X_t Y_t) = X_t dY_t + Y_t dX_t + d[X, Y]_t,$$

*where  $[X, Y]_t$  is the quadratic covariation of  $X_t$  and  $Y_t$ .*

**Proposition 1.** *The discrete-time solution to the Ornstein-Uhlenbeck SDE is given by*

$$X_{n+1} = e^{-\lambda \Delta t} X_n + \sigma \int_{t_n}^{t_{n+1}} e^{-\lambda(t_{n+1}-s)} dW_s.$$

*Proof.* Using the integrating factor  $e^{\lambda t}$ , a finite-variation process satisfying  $d[e^{\lambda t}, X_t] = 0$ , by Itô's product rule we have

$$\begin{aligned} d(e^{\lambda t} X_t) &= e^{\lambda t} dX_t + \lambda e^{\lambda t} X_t dt \\ &= e^{\lambda t} (-\lambda X_t dt + \sigma dW_t) + \lambda e^{\lambda t} X_t dt \\ &= e^{\lambda t} \sigma dW_t. \end{aligned}$$

Therefore, integrating from  $t_n$  to  $t_{n+1} = t_n + \Delta t$  gives

$$\begin{aligned} e^{\lambda t_{n+1}} X_{n+1} - e^{\lambda t_n} X_n &= \sigma \int_{t_n}^{t_{n+1}} e^{\lambda s} dW_s \\ X_{n+1} &= e^{-\lambda \Delta t} X_n + \sigma \int_{t_n}^{t_{n+1}} e^{-\lambda(t_{n+1}-s)} dW_s. \end{aligned}$$

□

## 2. DISTRIBUTION OF THE STOCHASTIC INTEGRAL

**Theorem 2.** (*Itô isometry*) Let  $W_t$  be a Brownian motion and let  $f \in L^2([a, b])$  be deterministic. Then

$$\mathbb{E} \left[ \left( \int_a^b f(s) dW_s \right)^2 \right] = \int_a^b f(s)^2 ds.$$

In particular,

$$\int_a^b f(s) dW_s \sim \mathcal{N} \left( 0, \int_a^b f(s)^2 ds \right).$$

**Corollary 1.** Let  $X_t$  satisfy the Ornstein-Uhlenbeck SDE. Then, for any timestep  $\Delta t > 0$ ,

$$X_{n+1} | X_n \sim \mathcal{N} \left( e^{-\lambda \Delta t} X_n, \sigma^2 \int_{t_n}^{t_{n+1}} e^{-2\lambda(t_{n+1}-s)} ds \right).$$

Equivalently,

$$X_{n+1} = e^{-\lambda \Delta t} X_n + \sigma \sqrt{\frac{1 - e^{-2\lambda \Delta t}}{2\lambda}} \xi_n, \quad \xi_n \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1) \quad (1)$$

*Proof.* By Proposition 1,

$$X_{n+1} = e^{-\lambda \Delta t} X_n + \sigma \int_{t_n}^{t_{n+1}} e^{-\lambda(t_{n+1}-s)} dW_s.$$

The stochastic integral is Gaussian with mean zero. By Itô isometry,

$$\begin{aligned} \text{Var} \left[ \int_{t_n}^{t_{n+1}} e^{-\lambda(t_{n+1}-s)} dW_s \right] &= \int_{t_n}^{t_{n+1}} e^{-2\lambda(t_{n+1}-s)} ds \\ &= \frac{1 - e^{-2\lambda \Delta t}}{2\lambda}. \end{aligned}$$

The result follows. □

Letting  $\lambda = 1/\tau_c$ , we obtain the form encountered in the main text:

$$X_{n+1} = e^{-\Delta t/\tau_c} X_n + \sigma \sqrt{\frac{\tau_c}{2} (1 - e^{-2\Delta t/\tau_c})} \xi_n.$$

Furthermore, note that the  $\xi_n$  are independent because the stochastic integrals are taken over disjoint Brownian intervals.

### 3. PREFERENCE OVER EULER-MARUYAMA

Since the Ornstein–Uhlenbeck process possesses a closed-form solution, its transition law may be sampled exactly. This preserves the prescribed variance and exponential correlation function for arbitrary timestep and avoids the timestep-dependent bias introduced by Euler–Maruyama discretization. The exact update is therefore both more accurate and computationally more efficient for the correlation times considered in this work.